

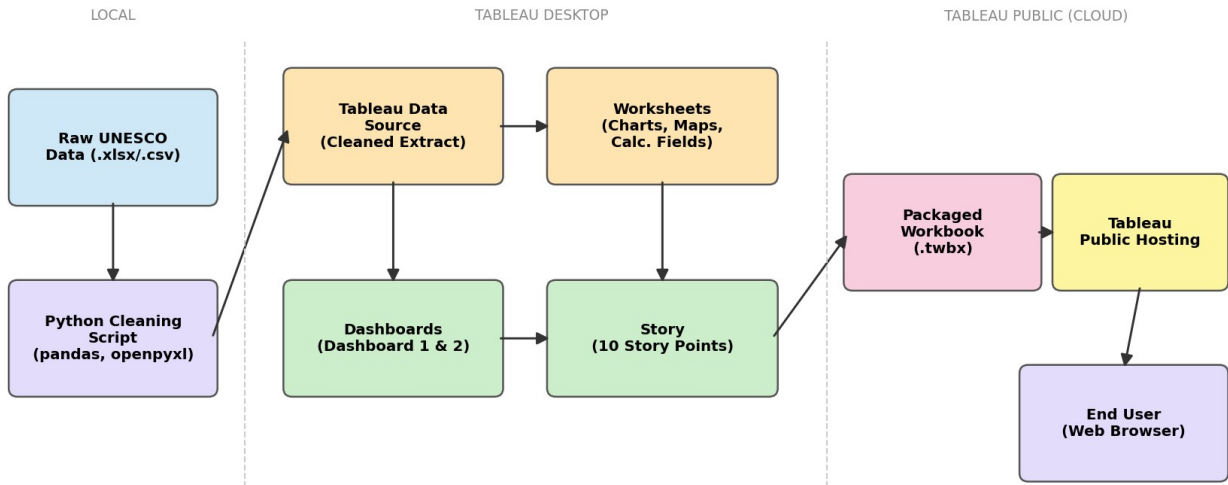
## Project Design Phase-II

### Technology Stack (Architecture & Stack)

|               |                                                                                    |
|---------------|------------------------------------------------------------------------------------|
| Date          | 06 July 2026                                                                       |
| Name          | Aditya Kumar Singh                                                                 |
| Project Name  | Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau |
| Maximum Marks | 4 Marks                                                                            |

#### Technical Architecture:

The deliverable below shows the architectural diagram for the Heritage Treasures UNESCO Tableau project, along with the information in Table 1 and Table 2.



**Table-1: Components & Technologies:**

| S.No | Component            | Description                                                           | Technology                                                    |
|------|----------------------|-----------------------------------------------------------------------|---------------------------------------------------------------|
| 1.   | User Interface       | How the viewer interacts with the dashboard                           | Tableau Public web viewer (browser-based, HTML5/JS rendering) |
| 2.   | Data Source          | Raw UNESCO World Heritage Sites dataset                               | Excel (.xlsx) — 1,121 rows × 18 columns                       |
| 3.   | Data Cleaning / ETL  | Removing HTML entities, whitespace, and duplicates; validating fields | Python (pandas, openpyxl)                                     |
| 4.   | Data Modeling        | Structuring cleaned data for visualization                            | Excel workbook: UNESCO_Clean_Data and Summary sheets          |
| 5.   | Visualization Engine | Building charts, dashboards, and the guided story                     | Tableau Desktop                                               |

|     |                                   |                                                    |                                                                                      |
|-----|-----------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------|
| 6.  | Publishing / Hosting              | Hosting the workbook for public access             | Tableau Public (cloud hosting)                                                       |
| 7.  | File Storage                      | Storage of source Excel file and Tableau workbook  | Local filesystem during development; Tableau Public cloud storage after publishing   |
| 8.  | External API                      | Map rendering service for the world map view       | Mapbox / OpenStreetMap (built into Tableau)                                          |
| 9.  | Machine Learning / Forecast Model | Forecasting future heritage site inscriptions      | Tableau built-in forecasting (exponential smoothing) on the Year Forecasting sheet   |
| 10. | Infrastructure                    | Application development and deployment environment | Local machine (Tableau Desktop) for development; Tableau Public cloud for deployment |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                          | Technology                                                                                    |
|------|--------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1.   | Open-Source Frameworks   | Libraries used for data cleaning and validation      | Python: pandas, openpyxl                                                                      |
| 2.   | Security Implementations | Data handling and access controls                    | Only public, non-personal UNESCO open data used; access limited to a read-only public URL     |
| 3.   | Scalable Architecture    | Justification for scalability of the workbook design | Tableau Data Extract (.hyper) refreshed as new UNESCO sites are inscribed each year           |
| 4.   | Availability             | Justification for application availability           | Hosted on Tableau Public's cloud infrastructure, available independent of the author's device |
| 5.   | Performance              | Design considerations for dashboard performance      | Tableau Hyper in-memory engine and pre-aggregated calculated fields for fast filter response  |

## References:

- <https://public.tableau.com/>
- <https://help.tableau.com/current/pro/desktop/en-us/>
- <https://whc.unesco.org/en/list/>
- <https://c4model.com/>